

Spring 2022

Applied quantitative Methods for Non-quantitative doctoral researchers

Session 08: Regression analysis I

Charles Tackney, **Jan Bauer** & Manuele Citi
Department of Management, Society & Communication
Copenhagen Business School

Regression analysis

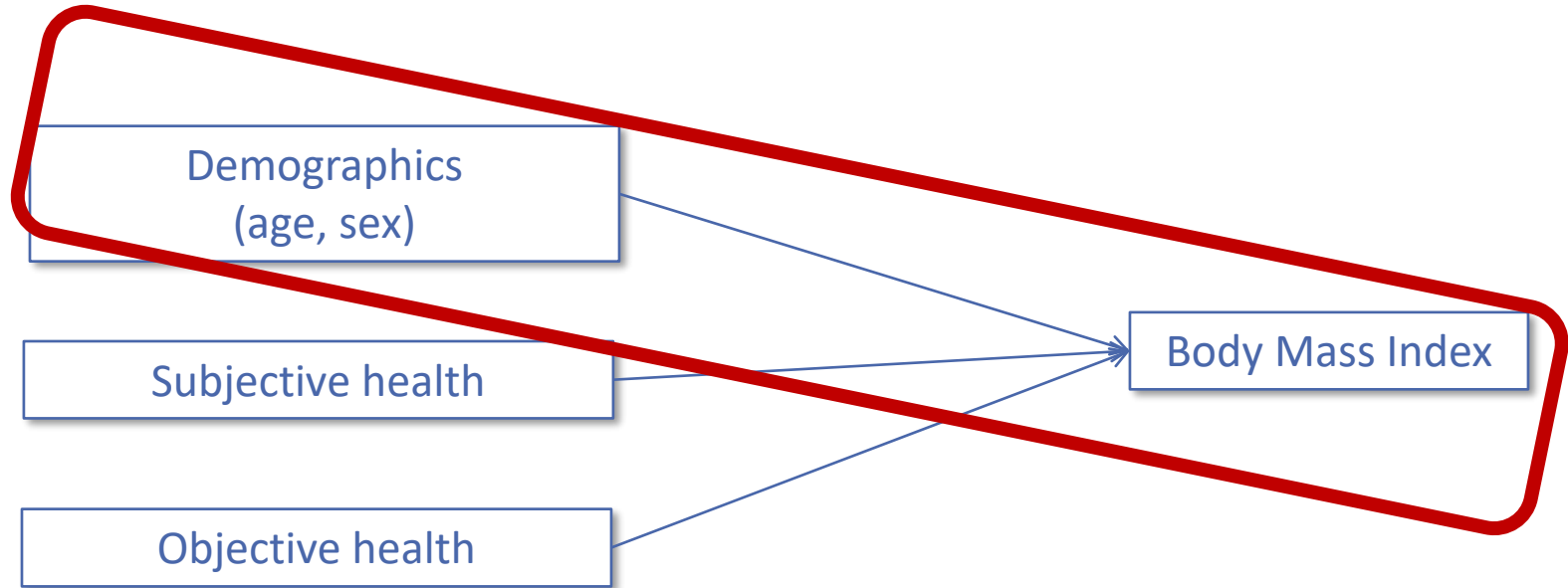
What linear regression is all about?

Analyze the relationship between
one **dependent variable (DV)** and
one or more **independent variable(s) (IV)**

Important:

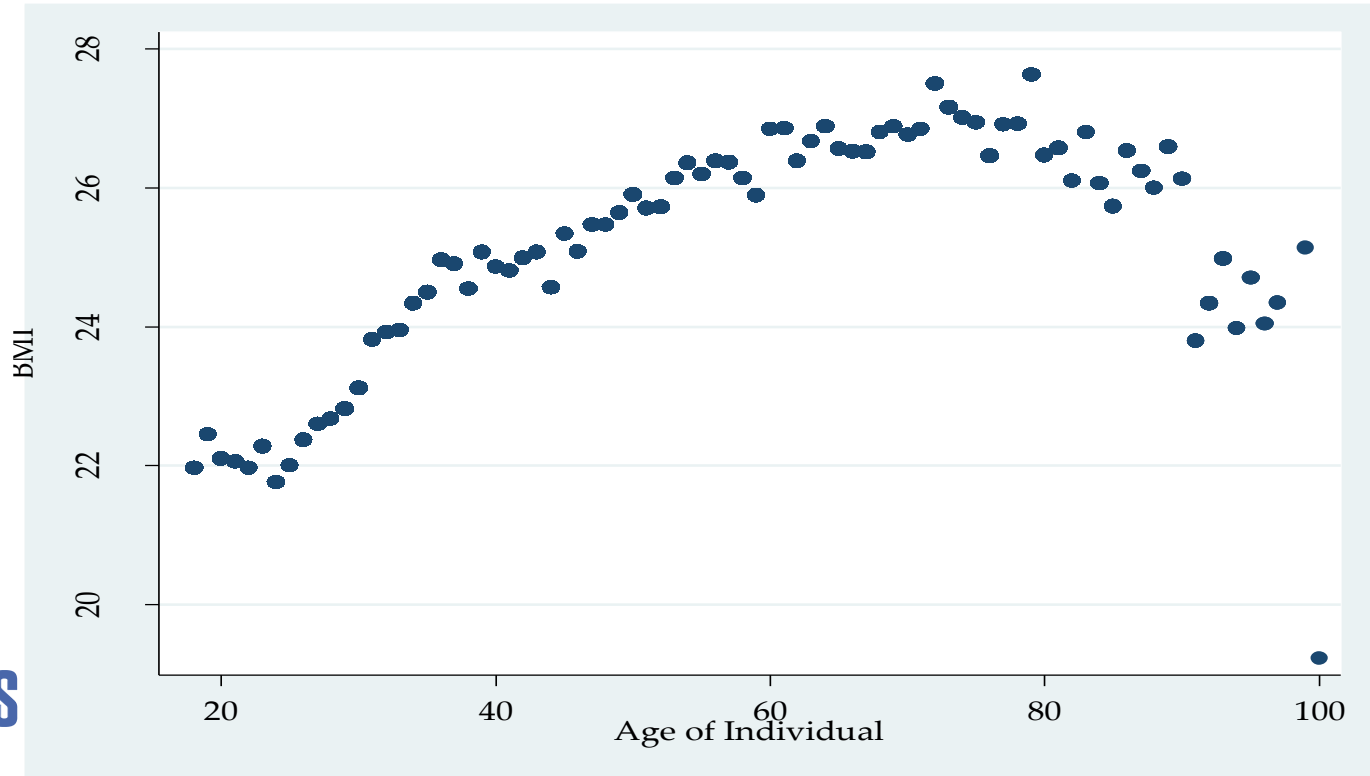
- The **DV** has to be in a metric scale
- The **IV** mustn't be nominal scaled (dummies are allowed)
- We measure “only” **linear** relationships
- The predicted relationship should be theory-based

An example



An example

Question: Is there a relationship between age and BMI?



Regression analysis – 3 steps

1. Formulate your model



2. Estimate regression equation

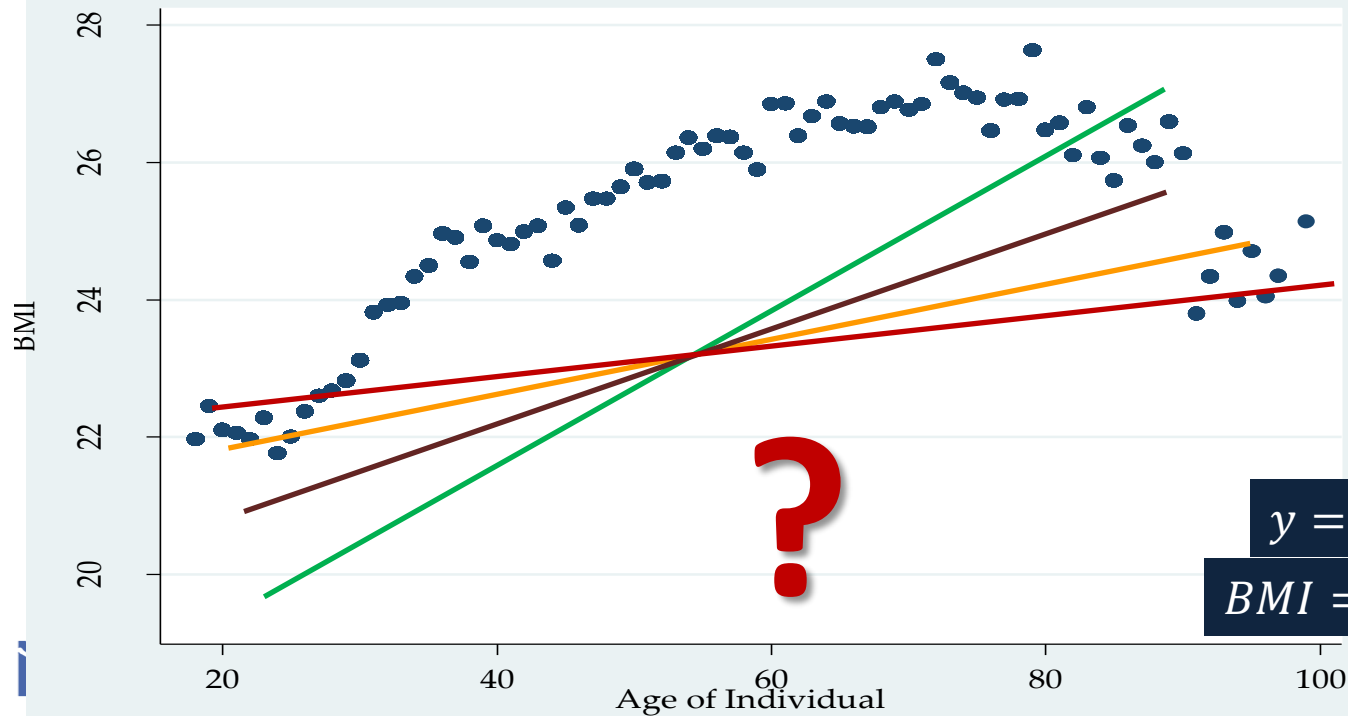


3. Test regression model

Regression analysis – 3 steps

Hypothesis: age is related to BMI

Can we describe the relationship by using the following line?

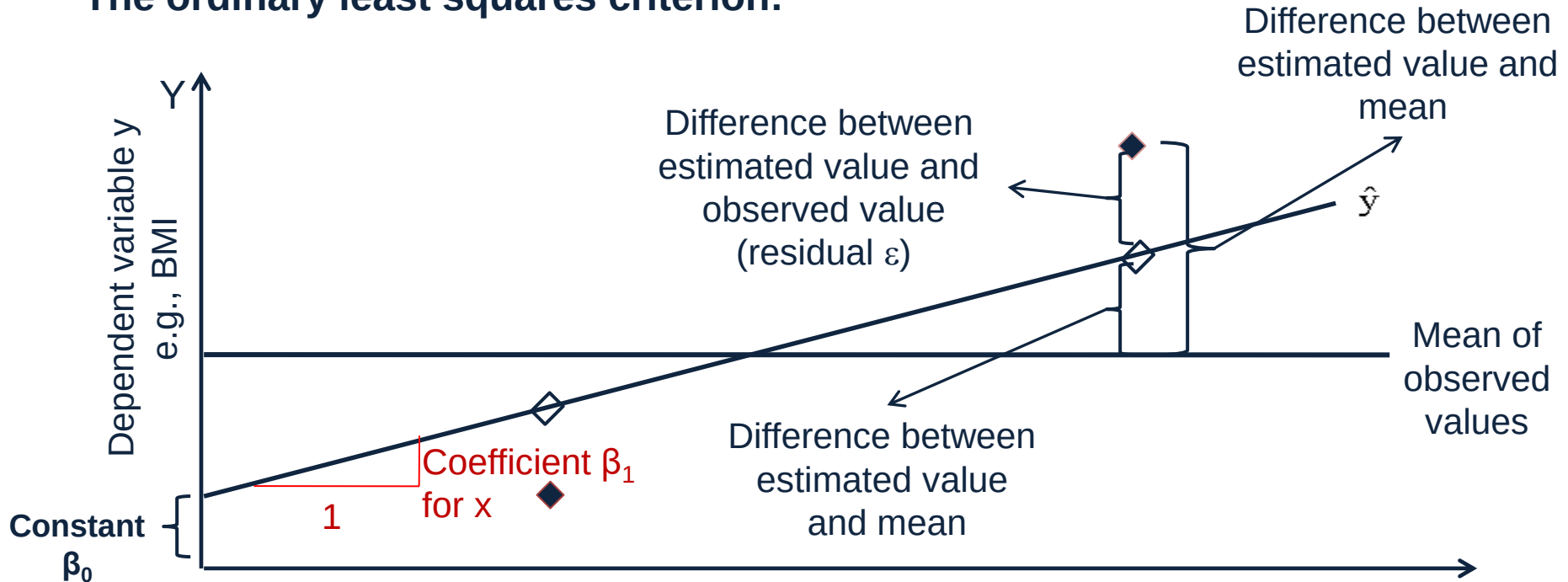


$$y = \beta_0 + \beta_1 x$$

$$BMI = \beta_0 + \beta_1 age$$

Regression analysis – 3 steps

The ordinary least squares criterion:



Regression analysis – 3 steps

OLS – is a mathematical procedure that generates a straight line that minimizes the sum of squared deviations of the actual values from the predicted regression line.

The resulting straight line will have the least possible total error in using X to predict Y .

Regression – Ordinary Least Squares (OLS)

$$\sum_{i=1}^n e_i^2 \text{ is minimum}$$

Regression analysis – 3 steps

The residual

Is the difference between the observed value and the estimated value = error

$$\varepsilon = y - (\beta_0 + \beta_1 x_1)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Least squares estimation

minimize the squared residuals by optimizing β_0 and β_1

Squared residuals

if you do not take the squared residuals,
residuals could sum up to zero (compensation)



Regression analysis – 3 steps

From our simplified example

```
. reg Y X
```

Source	SS	df	MS	Number of obs	=	3
Model	2	1	2	F(1, 1)	=	.
Residual	0	1	0	Prob > F	=	.
Total	2	2	1	R-squared	=	1.0000
				Adj R-squared	=	1.0000
				Root MSE	=	0

Y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
X	.5	.	.	.	.
_cons	5	.	.	.	.

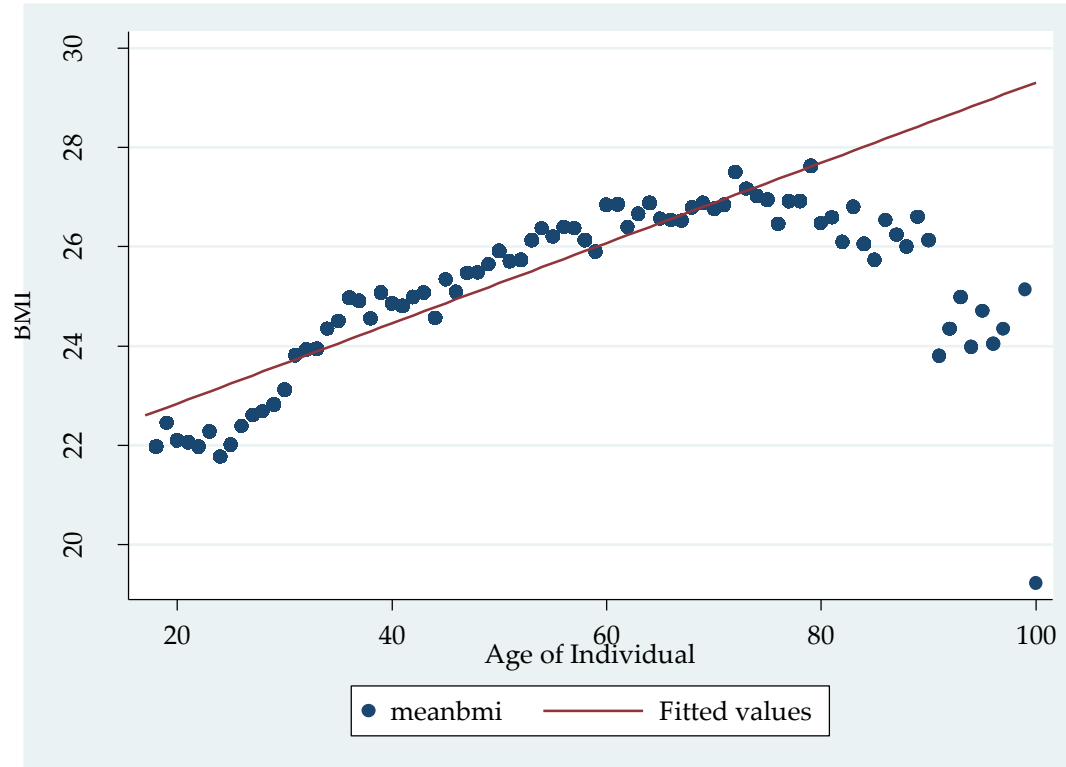
X	Y
12	11
10	10
8	9

$$Y = 5 + 0.5 * X$$

Regression analysis – 3 steps

Output from real data:

h_bmi	Coef.	Std. Err.
age	.0809353	.0017636
_cons	21.21955	.0937372



Regression analysis – 3 steps

```
. reg h_bmi age
```

Source	SS	df	MS	Number of obs	=	18,122
Model	37107.2831	1	37107.2831	F(1, 18120)	=	2106.07
Residual	319260.396	18,120	17.6192272	Prob > F	=	0.0000
Total	356367.679	18,121	19.6660052	R-squared	=	0.1041
				Adj R-squared	=	0.1041
				Root MSE	=	4.1975

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0809353	.0017636	45.89	0.000	.0774784	.0843921
_cons	21.21955	.0937372	226.37	0.000	21.03582	21.40329

Regression analysis – 3 steps

- a) **Is the regression model valid for population?**
 - R-square
 - F-test

- b) **What is the effect of the independent variables?**
 - Or is the effect just coincidental in the sample?
 - One sample t-test (hypothesis testing – 6 steps)
 - Confidence interval

Regression analysis – 3 steps

Number of obs = 18122

F(1, 18120) = 2106.07

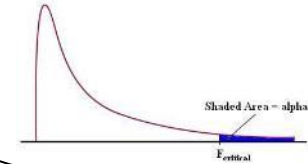
Prob > F = 0.0000

R-squared = 0.1041

Adj R-squared = 0.1041

Root MSE = 4.1975

F=2106
Prob > F: 0.000 < 0.05
It indicates that the
regression model is useful



About 10% of BMI's
variation is explained
by age

On average the observed
BMI is 4.19 points away
from the regression line vs.
4.43 from the simple mean

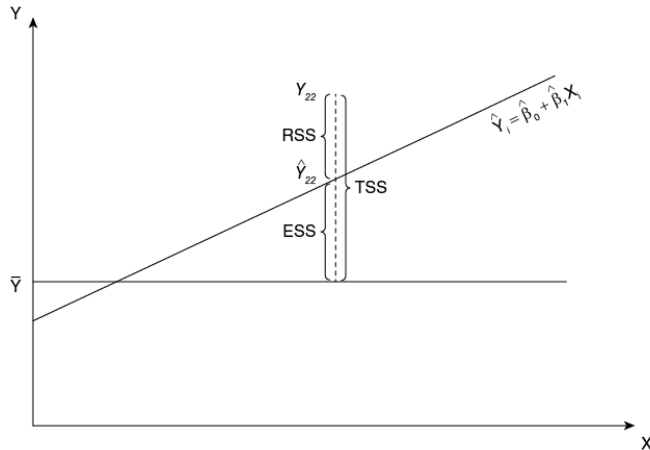
Regression analysis – 3 steps

h_bmi	Coef.	Std. Err.	t	P> t
age	.0809353	.0017636	45.89	0.000
_cons	21.21955	.0937372	226.37	0.000

[95% Conf. Interval]	
.0774784	.0843921
21.03582	21.40329

Sum of squares

Source	SS	df	MS
Model	37107.2831	1	37107.2831
Residual	319260.396	18,120	17.6192272
Total	356367.679	18,121	19.6660052



RSS = Residual SS

ESS = Model SS

TSS = Total SS

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y}_i)^2}{\sum (Y_i - \bar{Y}_i)^2}$$

$$= 37107/356367 = 0.1041$$

$$\begin{aligned} \text{Root MSE} &= \text{sqr}(\text{Residual MS}) \\ &= \text{sqr}(17.61) \\ &= 4.19 \end{aligned}$$

Standardized coefficients

```
. reg h_bmi age,beta
```

Source	SS	df	MS	Number of obs	=	18,122
Model	37107.2831	1	37107.2831	F(1, 18120)	=	2106.07
Residual	319260.396	18,120	17.6192272	Prob > F	=	0.0000
Total	356367.679	18,121	19.6660052	R-squared	=	0.1041
				Adj R-squared	=	0.1041
				Root MSE	=	4.1975

h_bmi	Coef.	Std. Err.	t	P> t	Beta
age	.0809353	.0017636	45.89	0.000	.3226862
_cons	21.21955	.0937372	226.37	0.000	.

CBS



COPENHAGEN BUSINESS SCHOOL
HANDELSHØJSKOLEN

18

Binary independent variables (dummy regression)

```
. tab mstat
```

Marital status	Freq.	Percent	Cum.
Married, Living Together	12,159	59.09	59.09
Separated (still married)	366	1.78	60.87
Single	4,951	24.06	84.93
Divorced	1,650	8.02	92.95
Widowed	1,451	7.05	100.00
Total	20,577	100.00	

```
. reg h_bmi mstat
```

Source	SS	df	MS	Number of obs	=	18,121
Model	2320.95122	1	2320.95122	F(1, 18119)	=	118.78
Residual	354045.36	18,119	19.5400055	Prob > F	=	0.0000
Total	356366.311	18,120	19.667015	R-squared	=	0.0065
				Adj R-squared	=	0.0065
				Root MSE	=	4.4204

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
mstat	-.2672987	.024526	-10.90	0.000	-.3153719 -.2192255
_cons	25.81475	.0593238	435.15	0.000	25.69847 25.93103

```
. tab mstat,nol
```

Marital status	Freq.	Percent	Cum.
1	12,159	59.09	59.09
2	366	1.78	60.87
3	4,951	24.06	84.93
4	1,650	8.02	92.95
5	1,451	7.05	100.00
Total	20,577	100.00	

RQ: Does marital status affect BMI?

```
. reg h_bmi i.mstat
```

Source	SS	df	MS	Number of obs	=	18,121
Model	26079.624	4	6519.90599	F(4, 18116)	=	357.61
Residual	330286.687	18,116	18.2317668	Prob > F	=	0.0000
				R-squared	=	0.0732
				Adj R-squared	=	0.0730
Total	356366.311	18,120	19.667015	Root MSE	=	4.2699

	h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
mstat						
Separated (still married)		-.4221939	.2433183	-1.74	0.083	-.8991208 .054733
Single		-2.682856	.0771045	-34.80	0.000	-2.833988 -2.531724
Divorced		-.5162922	.1199439	-4.30	0.000	-.7513936 -.2811908
Widowed		1.0632	.1254878	8.47	0.000	.8172317 1.309168
_cons		25.8829	.0411135	629.55	0.000	25.80231 25.96348

```
. reg h_bmi ib3.mstat
```

Source	SS	df	MS	Number of obs	=	18,121
Model	26079.624	4	6519.90599	F(4, 18116)	=	357.61
Residual	330286.687	18,116	18.2317668	Prob > F	=	0.0000
				R-squared	=	0.0732
				Adj R-squared	=	0.0730
Total	356366.311	18,120	19.667015	Root MSE	=	4.2699

	h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
mstat						
Married, Living Together		2.682856	.0771045	34.80	0.000	2.531724 2.833988
Separated (still married)		2.260662	.2485322	9.10	0.000	1.773515 2.747809
Divorced		2.166564	.130196	16.64	0.000	1.911367 2.42176
Widowed		3.746056	.1353206	27.68	0.000	3.480814 4.011297
_cons		23.20004	.0652287	355.67	0.000	23.07219 23.3279

dummy regression

$$bmi_i = \beta_0 + \beta_1 * \text{Separated} + \beta_2 * \text{Single} + \beta_3 * \text{Divorced} + \beta_4 * \text{Widowed} + e_i$$

Nominal variables have to be recoded to a set of dummies and can only be interpreted against one baseline/reference.

Married	Separated	Single	Divorced	Widowed
0	0	0	0	0
1	1	1	1	1

Predicted values and graphs

```
. reg h_bmi age
```

Source	SS	df	MS	Number of obs	=	18,122
				F(1, 18120)	=	2106.07
Model	37107.2831	1	37107.2831	Prob > F	=	0.0000
Residual	319260.396	18,120	17.6192272	R-squared	=	0.1041
				Adj R-squared	=	0.1041
Total	356367.679	18,121	19.6660052	Root MSE	=	4.1975

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0809353	.0017636	45.89	0.000	.0774784	.0843921
_cons	21.21955	.0937372	226.37	0.000	21.03582	21.40329

```
. margins, atmeans
```

```
Adjusted predictions      Number of obs      =      18,122
Model VCE      : OLS
```

```
Expression      : Linear prediction, predict()
at              : age              =      50.12405 (mean)
```

	Delta-method					
	Margin	Std. Err.	t	P> t	[95% Conf. Interval]	
_cons	25.27636	.031181	810.63	0.000	25.21524	25.33747

STATA has the option directly calculate and graph predicted values from the regression using the *margins* command.

$$\text{Bmi}_i = \beta_0 + \beta_1 * \text{age}_i$$

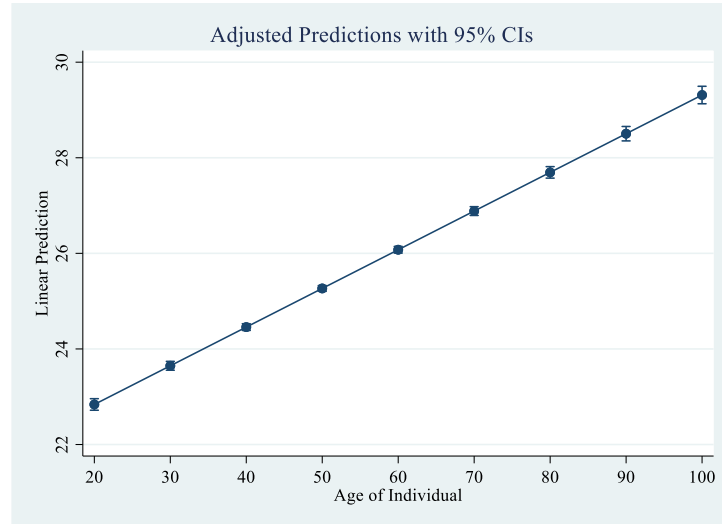
$$\begin{aligned} \text{Bmi}_i &= 21.21 + 0.08 * 50.12 \\ &= 25.27 \end{aligned}$$

Predicted values and graphs

```
. qui: reg h_bmi age
```

```
. margins, at(age=(20(10)100))
```

```
. marginsplot
```



Expression : Linear prediction, predict()

1._at	: age	=	20
2._at	: age	=	30
3._at	: age	=	40
4._at	: age	=	50
5._at	: age	=	60
6._at	: age	=	70
7._at	: age	=	80
8._at	: age	=	90
9._at	: age	=	100

	Delta-method				
	Margin	Std. Err.	t	P> t	[95% Conf. Interval]
_at					
1	22.83826	.0616014	370.74	0.000	22.71751 22.959
2	23.64761	.0472426	500.56	0.000	23.55501 23.74021
3	24.45696	.0359312	680.66	0.000	24.38654 24.52739
4	25.26632	.0311818	810.29	0.000	25.2052 25.32744
5	26.07567	.0357158	730.09	0.000	26.00566 26.14568
6	26.88502	.0469148	573.06	0.000	26.79306 26.97698
7	27.69437	.0612244	452.34	0.000	27.57437 27.81438
8	28.50373	.0769281	370.52	0.000	28.35294 28.65451
9	29.31308	.0933246	314.10	0.000	29.13015 29.496

Application

Let's try this.....